Partial Fractions with Linear Denominators

Exam code: 9MA0

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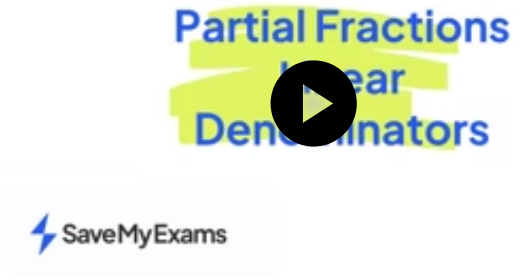
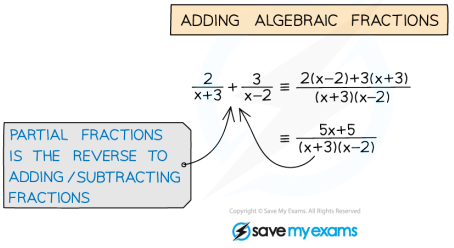


Did this video help you? Yes No

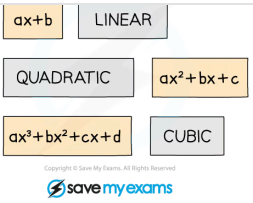
Partial fractions with linear denominators

What are partial fractions?

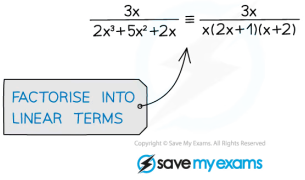


- This is the reverse process to **adding** (or subtracting) fractions
- When adding fractions a common denominator is required
- In partial fractions the common denominator is split into parts (factors)
- Partial fractions are used in **binomial expansions** (see Multiple GBEs) and **integration** (see Integration by Partial Fractions)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- A **linear factor** is of the form (**ax + b**)
- A **non-linear denominator** may be written as the **product** of linear factors
- If the denominator can be **factorised**



How do I find partial fractions?

STEP 1 **Factorise** the **polynomial** in the **denominator**

(Sometimes the numerator can be factorised too)

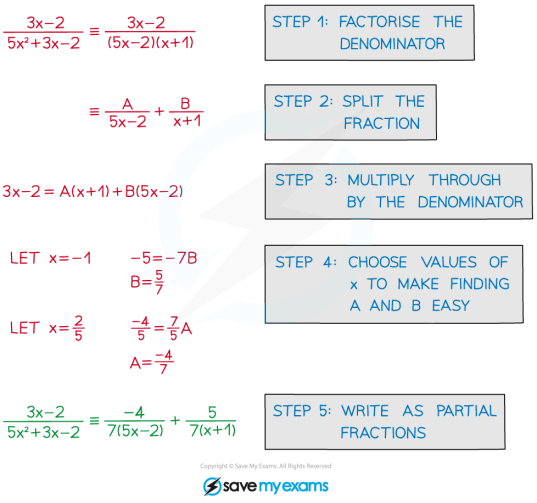
STEP 2 **Split** the fraction into a **sum** with **single** linear denominators

STEP 3 Multiply by the denominator to get rid of fractions

STEP 4 **Substitute** values of **x** to find **A**, **B**, etc

(An alternative method is **comparing coefficients**)

STEP 5 Write the original as partial fractions



- "The number of $\mathbf{x^2}$ on the LHS" = "The number of $\mathbf{x^2}$ on the RHS"
- "The number of ..." is called the **coefficient** of x^2

$3x-2 = A(x+1) + B(5x-2)$

COMPARE COEFFICIENTS OF x: $3=A+5B$ (•)

COMPARE CONSTANT TERMS: $-2=A-2B$ (••)

STEP 4 ALTERNATIVE: COMPARING COEFFICIENTS

(•) − (••): $5=7B$ $\therefore B=\frac{5}{7}$

$3=A+\frac{25}{7}$ $\therefore A=\frac{-4}{7}$

(•) & (••) CAN BE SOLVED SIMULTANEOUSLY

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Worked Example

Express $\dfrac{2x+5}{3x^3-12x}$ in partial fractions.

$$\frac{2x+5}{3x^3-12x} \equiv \frac{2x+5}{3x(x^2-4)}$$

$$\equiv \frac{2x+5}{3x(x+2)(x-2)}$$

STEP 1: FACTORISE THE DENOMINATOR

STEP 2: SPLIT THE FRACTION

$$\frac{2x+5}{3x(x+2)(x-2)} \equiv \frac{A}{3x} + \frac{B}{x+2} + \frac{C}{x-2}$$

STEP 3: GET RID OF FRACTIONS

$\therefore\ 2x+5 \equiv A(x+2)(x-2) + B(3x)(x-2) + C(3x)(x+2)$

LET $x=0$: $5=-4A$ $\therefore A=\frac{-5}{4}$

LET $x=2$: $9=24C$ $\therefore C=\frac{9}{24}=\frac{3}{8}$

LET $x=-2$: $1=24B$ $\therefore B=\frac{1}{24}$

STEP 4: CHOOSE SUITABLE VALUES OF x

STEP 5: WRITE ORIGINAL FRACTION AS PARTIAL FRACTIONS

$$\therefore \frac{2x+5}{3x^3-12x} \equiv \frac{-5}{12x} + \frac{1}{24(x+2)} + \frac{3}{8(x-2)}$$

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